

PetPal AI — FBLA Introduction to Programming 2025-2026

Developer: Tharun Naguleswaran **Event:** Introduction to Programming **Technologies:** HTML5, CSS3, Vanilla JavaScript (ES6+), Chart.js, Groq API

2025-2026 Topic Correlation

Topic: *Build a Virtual Pet* — Create a digital pet that users can name, feed, and care for over time. The pet should respond to how well it is treated, showing emotions, changing appearance or behavior, and developing based on the user's actions. Include a "cost of care" system to teach users about the financial responsibility of pet ownership.

The table below maps every requirement to the specific feature in PetPal AI that addresses it.

Topic Requirement	How PetPal Addresses It
User names the pet, chooses type	3-step setup wizard — name (validated), pet type (Dog/Cat/Rabbit/Bird), color, and personality
Feed, play, rest, clean, health check	Five care actions: Feed, Play, Sleep, Clean, Vet — each with real coin costs and stat effects
Reactions based on care level	8 distinct emotions (ecstatic, happy, neutral, sad, hungry, tired, sick, dirty) driven live by stat values
Changing appearance or behavior	Pixel sprite renderer updates the pet's displayed frame based on current emotion and action state
Developing based on user's actions	XP and leveling system (10 levels); personality traits shift over time based on how often the user feeds vs. plays
Food and supply costs	Feed costs 15 coins, Play costs 10 coins, Clean costs 12 coins
Vet visits or health care	Vet costs 35 coins; sickness mechanic forces urgent vet visits
Budget limits or in-game currency	Coin system with a starting balance; actions are disabled when the player cannot afford them
Running total of care-related expenses	Finance tab shows total spent, spending by category, daily average, and monthly bar chart
Optional — earning system	6 chores (Wash Dishes, Clean Room, Water Plants, etc.) earn coins and XP on a cooldown timer
Optional — learn tricks	9 tricks unlock automatically as the pet reaches each level milestone
Optional — earn badges	12 badges unlock when specific milestones are reached (e.g. 10 feeds, happy streak, max level)

How to Run

1. Open `index.html` in any modern browser (Chrome, Firefox, Safari, Edge)
2. No installation or internet connection required for core gameplay
3. Complete the 3-step setup wizard to create your pet
4. Use the tabs (Care, Finance, Growth, Chores, History, Memories, Chat, Reports) to manage and analyze your pet

How to Use

Setup Wizard

- **Step 1** — Choose a pet type (Dog, Cat, Rabbit, or Bird)
- **Step 2** — Choose a color and personality (each personality has a unique gameplay effect)
- **Step 3** — Name your pet (2–20 characters, letters/numbers/hyphens/apostrophes only)

Care Actions

Action	Cost	Primary Effect
Feed	15 coins	Restores hunger; small happiness and health boost
Play	10 coins	Raises happiness; costs energy and hunger
Sleep	Free	Restores energy; small health boost
Clean	12 coins	Restores cleanliness; health and happiness boost
Vet	35 coins	Restores health; cures sickness

Keyboard Shortcuts

Key	Action
F	Feed
P	Play
S	Sleep
C	Clean
V	Vet
Ctrl+S	Save game
?	Open help

Tabs

- **Care** — Main dashboard with pet stats and action buttons
- **Finance** — Spending breakdown by category and transaction history
- **Growth** — XP progress, badges, tricks, and personality traits

- **Chores** — Earn coins and XP by completing timed chores
- **History** — Timestamped activity log
- **Memories** — Auto-recorded scrapbook of milestone moments
- **Chat** — AI advisor (Dr. Paws) powered by Groq — requires a free Groq API key
- **Reports** — Chart.js analytics with date range filter and downloadable reports

Reports

Four report types are available, each with a date range filter (7 / 30 / 90 days / All Time) and a Save button that downloads a formatted `.txt` file:

- **Expense Report** — spending by category (doughnut chart) and monthly totals (bar chart)
- **Health Report** — line chart of all 5 stats over time; toggle individual stats on/off
- **Achievement Report** — badge and trick progress (stacked bar); XP earned in the selected period
- **Care History Report** — count of each care action performed in the selected period

Project Structure

```
fbla/
├── index.html           Main application (single-page)
├── css/
│   ├── styles.css      Main stylesheet
│   └── animations.css   Keyframe animations
├── js/
│   ├── constants.js    All game config, costs, effects, badge/trick/chore
│   └── definitions
│       ├── icons.js     Pixel SVG icon library
│       ├── PixelSprites.js Animated pixel art pet renderer
│       ├── Validator.js  Centralized input validation (syntactical + semantic)
│       ├── Pet.js        Pet model – stats, XP, leveling, tick decay
│       ├── StorageManager.js localStorage persistence (game, expenses, income)
│       ├── Game.js       Game coordinator – actions, chores, badges, save/load
│       ├── AICompanion.js Groq streaming chat integration
│       ├── PersonalityEngine.js Adaptive personality trait system
│       ├── Scrapbook.js  Auto-recorded milestone memory journal
│       ├── ReportsCenter.js Chart.js analytics with date filtering and downloads
│       ├── HelpCenter.js Searchable FAQ with 20+ entries
│       ├── UIController.js All DOM interaction and game loop
│       └── main.js       Application entry point
├── docs/               Additional technical documentation
├── README.md           This file
└── RESOURCES.md        Libraries and attribution
```

Additional Features

Feature	Description
Personality Engine	Each personality (Friendly, Energetic, Shy, Wise, Funny) permanently changes how stats are affected, creating different gameplay strategies
Sickness mechanic	3% random chance of illness when health drops below 40; sick pets decay faster and are restricted to vet/sleep only
Memory Scrapbook	Automatically records milestone moments (level-ups, badges, days together, spending milestones) as journal entries
AI Chat (Dr. Paws)	Groq-powered streaming chat advisor that gives context-aware advice based on the pet's current stats
Stat toggle reports	Health report lets the user show/hide individual stat lines on the chart
Downloadable reports	All four reports export formatted <code>.txt</code> files

Browser Compatibility

Chrome 90+, Firefox 88+, Safari 14+, Edge 90+

Credits and Attribution

See [RESOURCES.md](#) for the complete list of external libraries and resources used.